Reverb

Intro to Audio Tech 1
Jocelyn Kavanagh
Wednesday, November 12

Georgia Tech

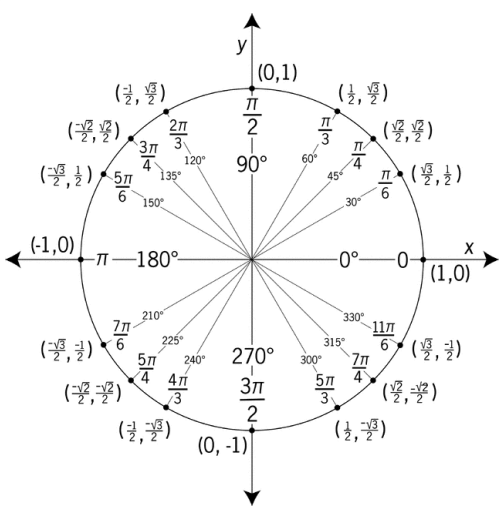
Lesson Outline

- Reverb
- HW5
- Quiz 3
- Up Next…

Checkpoint 10

- A complex number has magnitude one and argument pi/4. What are its real and imaginary parts?

This is polar form; we need to rewrite in cartesian/rectangular form. So…



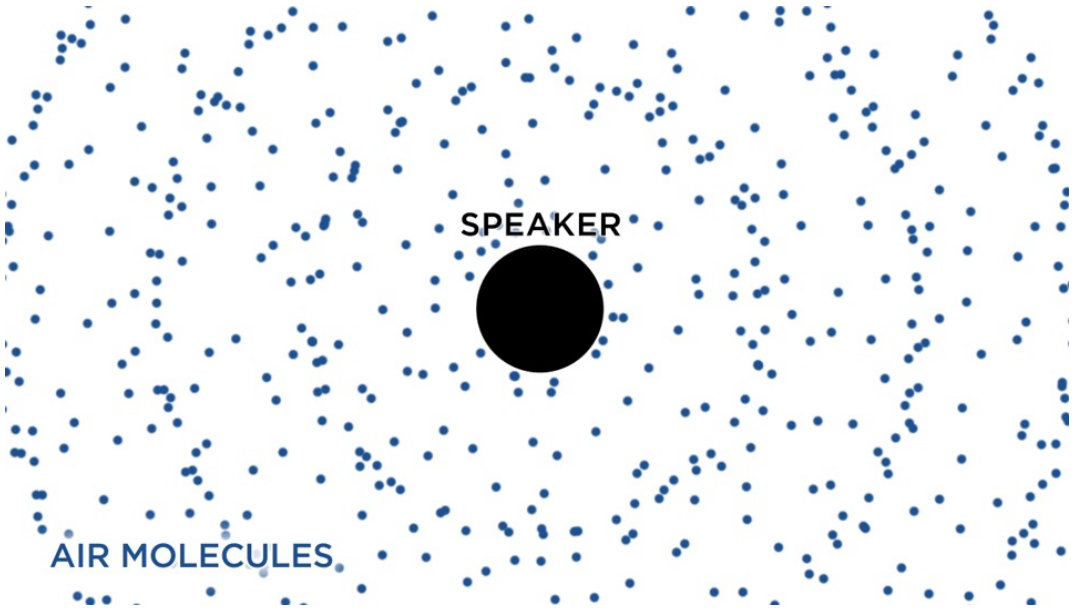
X (real component) = 1 * cos(pi/4) =

Y (imag component) = 1j * sin(pi/4)

We get $\frac{\sqrt{2}}{2} + j\,\frac{\sqrt{2}}{2}$

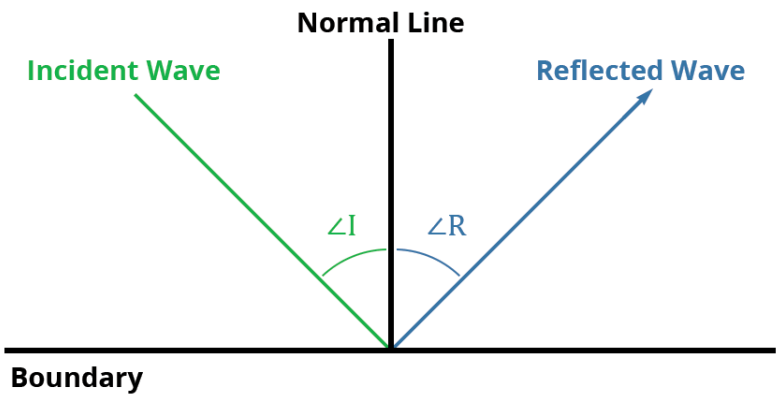
Reverb

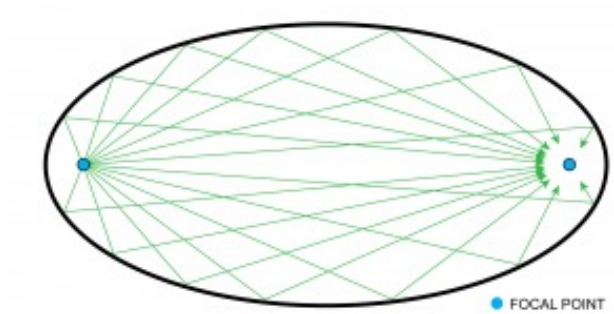
Propagation



Reflection and Transmission

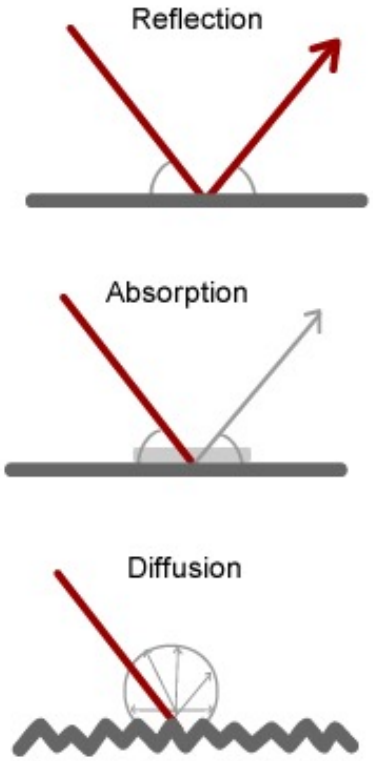
- In the real world, sound will be
  - Transmitted into a new medium (absorbed)
  - Reflected back into the original medium
- Law of reflection
- Diffraction and Refraction
  - Bending the sound waves based on obstacles or medium

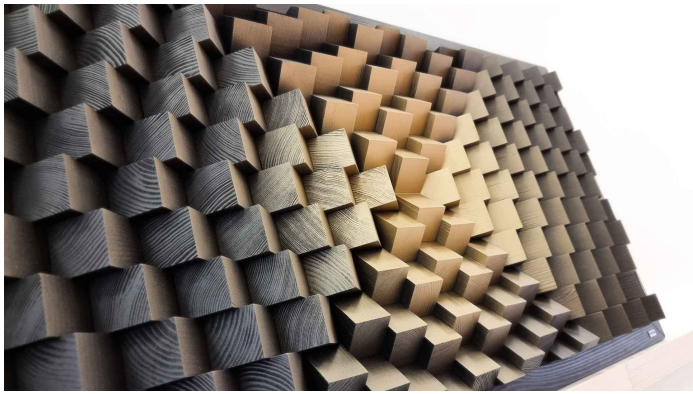




Diffusion

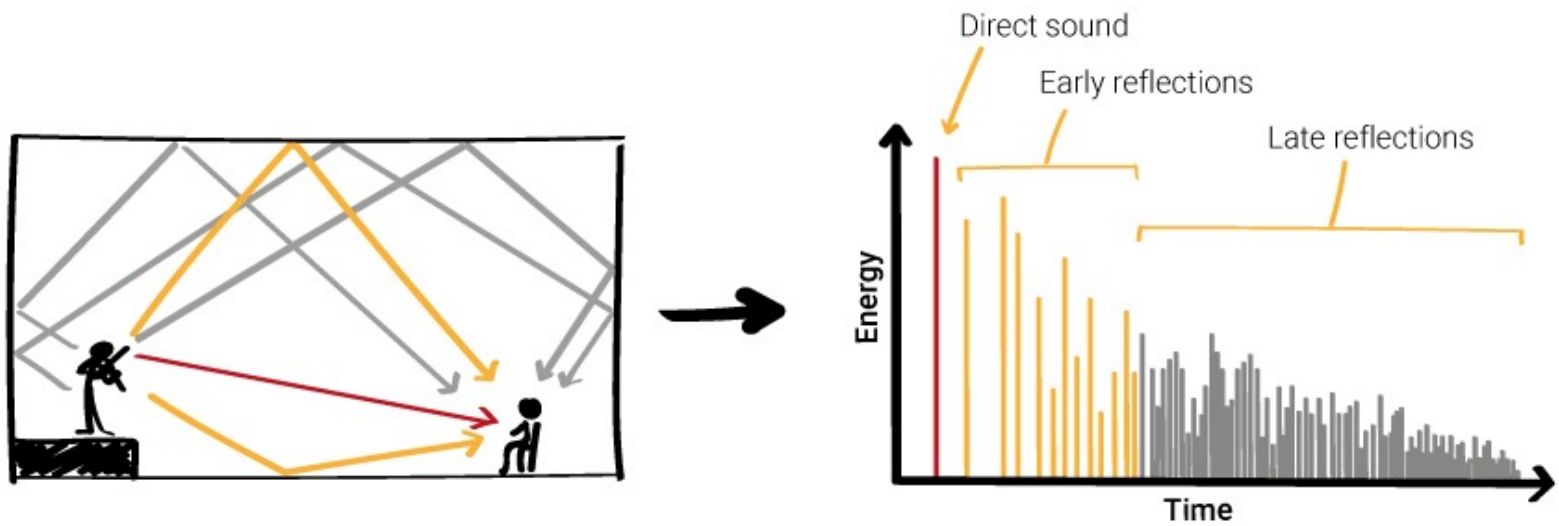
- Relates to reflection, but aims to distribute the sound more evenly in a space
- Curves and bends can help redistribute the sound
- Diffusers are used to treat rooms to give them a more ”natural” response





Reverberation

- The result of the reflecting sound waves
- Direct signal -> source to ear
- Early reflections -> source to “wall” to ear
- Late reflections -> source to “wall” to ”wall” etc
- Echoes are reflections that take around 100 ms to get to the ear



Impulse Response

- Measure of a room's response over time
- Sum of filtered and delayed reflections
- Properties of Impulse Response
  - Level decreases linearly
  - Density of the reflections increases
- Reverberation Time
  - Time in seconds for a level decrease of 60 dB – RT60
  - Depends on room volume, surface area, surface absorption
  - Sabine Equation:

$$T_{rt} = 0.163m^{-1}\frac{V}{\sum q_n S_n}$$

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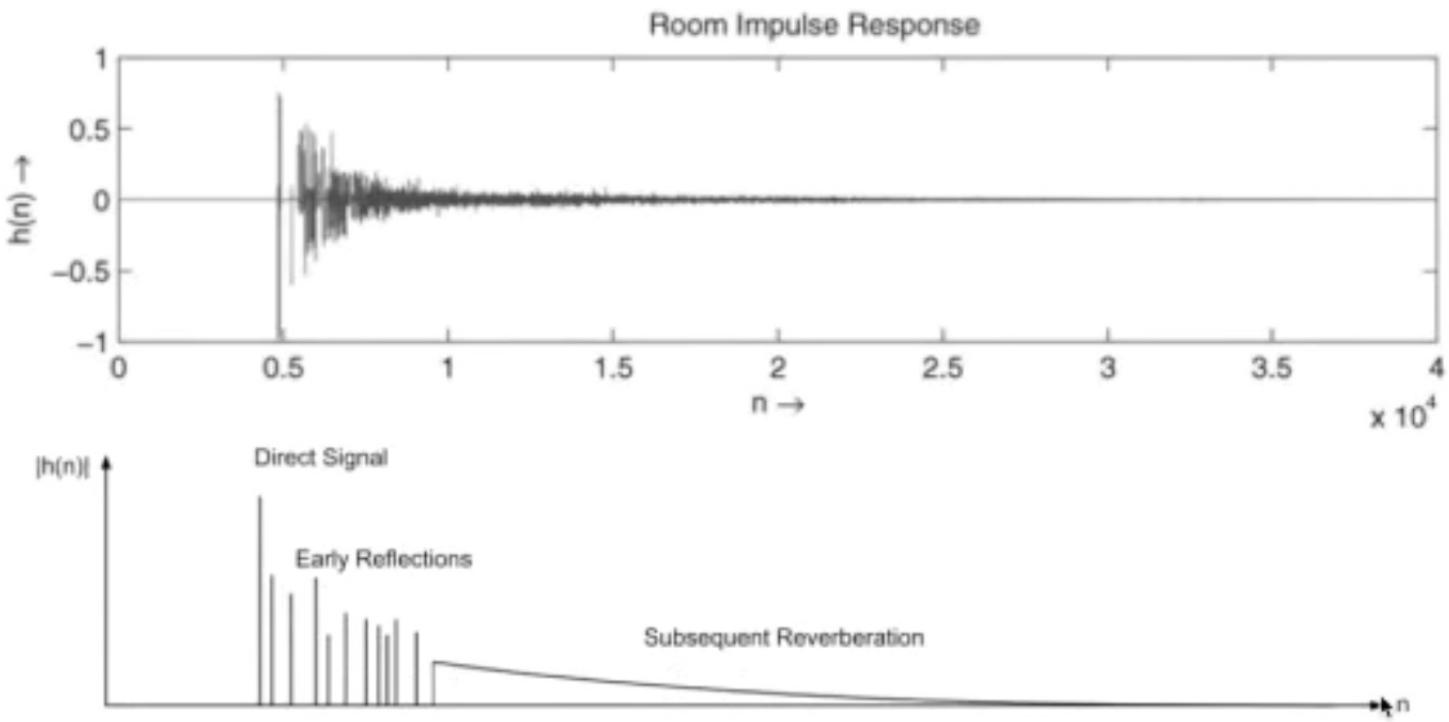
Impulse Response



Artificial Reverb

- Generate the impression of envelopment and reverberation
- Allow us to modify specific characteristics to “model” a room
- Approaches
  - Physical
    - Echo chambers, plate reverb, spring reverb
  - Digital
    - Spatial Modeling (raytracing)
    - Convolutoin
    - Digital/Algorithmic Reverberators – unit reverberators

Reverb Params

- Pre-delay
  - Amount of time between direct sound and early reflections
  - Room size
- Decay time / Reverb Time / Tail / RT60
  - Amount of time it takes to decrease the reverb by 60 dB
  - Room materials
- Diffusion
  - Density of reflections
  - Less density/low diffusion means clearer reflections
- Damping
  - The amount to cut high frequencies
  - High damping means more high frequencies cut, usually a darker sound
- Mix or Dry/Wet
  - the amount of unaffected vs affected signal

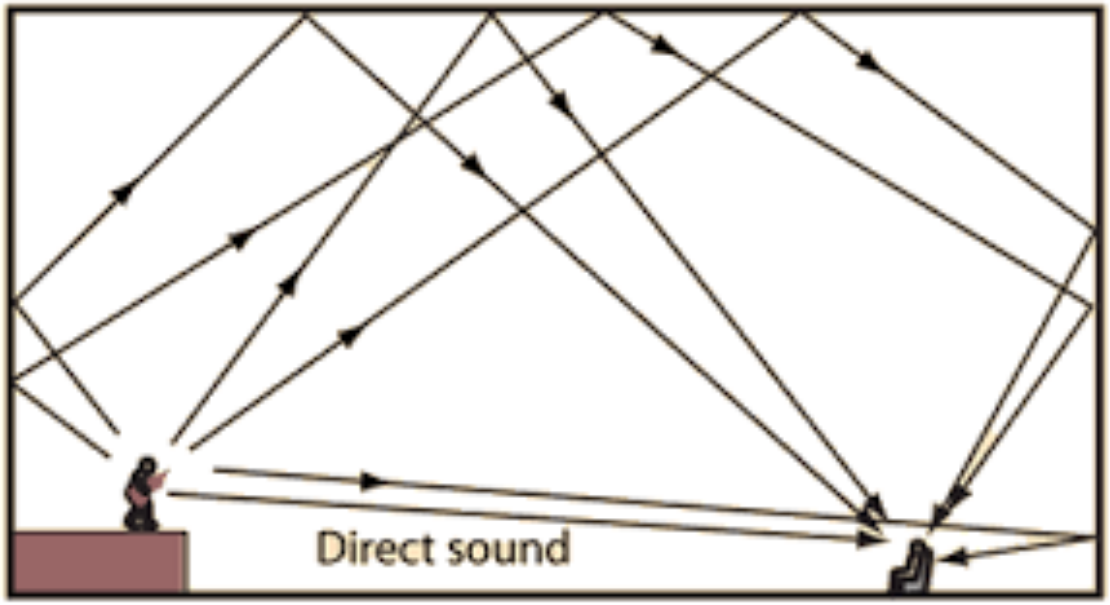
Analog Reverb

- Reverb Chamber/Room
  - Specially built rooms with high reverberation times
  - Speakers placed at one end and a mic in another
- Spring Reverb
  - Transducer sends sound down a spring which is connected to another transducer
- Plate Reverb
  - Transducer sends sound to a large, spring-suspended metal plate which is connected to another transducer

Reverb Chambers

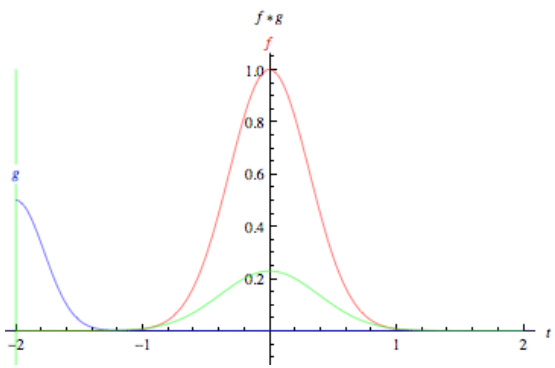
Creating Reverb: Spatial Modeling/Simulation

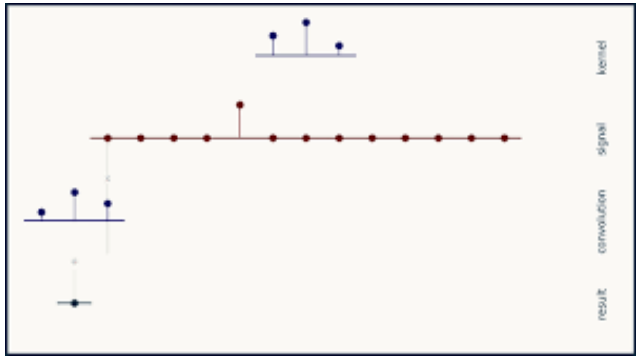
- Ray Tracing



Convolution

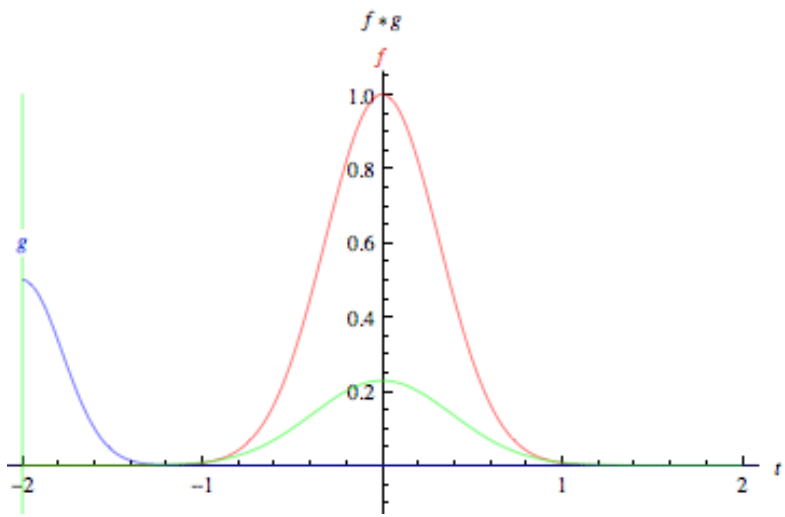
- Expresses how the “shape” of one function modifies the other
  - An integral that expresses the amount of overlap of one function as its shifted over another
  - ”Blends” two functions
- $(f * g)(t)$

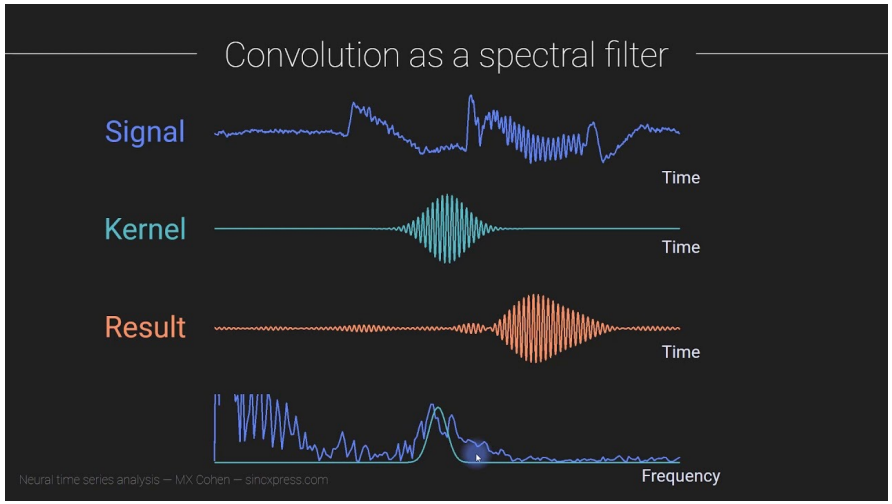




Fast Convolution and Reverb

- Take the impulse response of a room and convolve it with our sound
- This treats the reverb effect like a filter and “copies” the spectral properties of the impulse response onto our sound



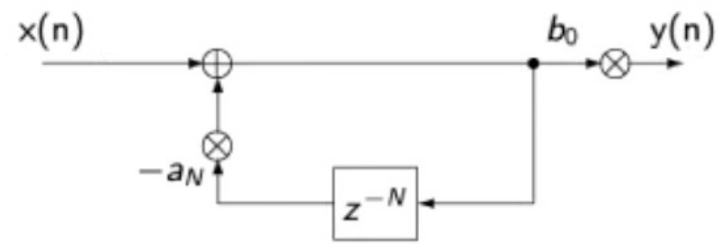


Algorithmic Reverb and Unit Reverberators

- Manfred Schroeder was the first person to implement digital artificial reverb at Bell Labs in the 1960s
- For the reverb effect we need:
  - The direct sound
  - Early reflections
  - Fused reverberation
- We can use a series of delays and tapped delay lines to create the reverberation effect
- Schroeder called the base components “unit reverberators” comprised of feedback comb filters and allpass filters

IIR Comb Filter

- A comb filter is a feedback loop with a scaled delayed signal continuously routed back to the input
- At short delays, it creates a “comb-like” spectral effect
- At long delays, it creates a series of echoes

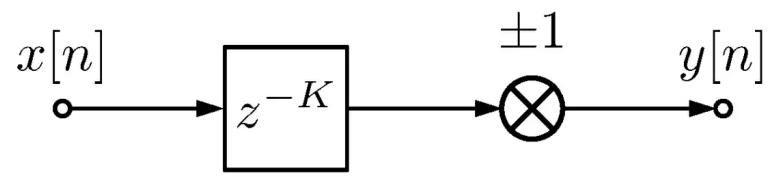


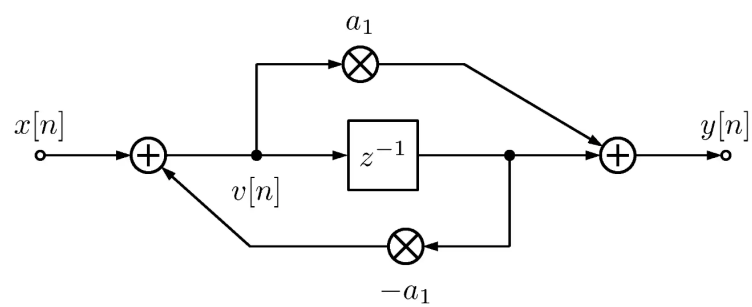
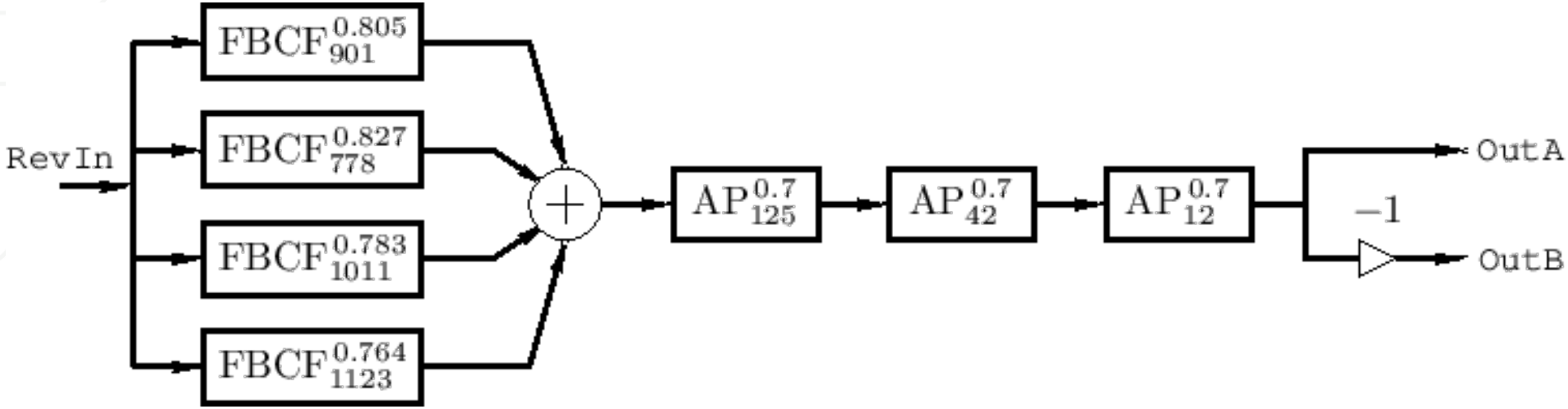
$$y(n) = b_0 \cdot x(n) - a_N \cdot y(n-N)$$

$$H(z) = \frac{b_0}{1 - a_N \cdot z^{-N}}$$

Allpass Filter

- An allpass filter generates a flat frequency response, passes all frequencies equally well
- Allpass filters cause a coloration of the sound through frequency-dependent phase shifting
- An FIR allpass is just a delay
- IIR allpass filters work similarly to the comb, but they cancel out the spectral effect through subtraction





Schroeder Reverb